# ECON2103 Microeconomics

## Chapter 10 Market Power: Monopoly and Monopsony

Dr. Sherry Zhou



BEIJING NORMAL UNIVERSITY · HONG KONG BAPTIST UNIVERSITY
UNITED INTERNATIONAL COLLEGE

# Outline

1. Monopoly
2. Monopoly Power
3. Sources of Monopoly Power
4. The Social Costs of Monopoly Power
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7. Summary and Exercises

# Market Power: Nonopoly and Monopsony

垄断

- **monopoly** - Market with only one seller.
- **monopsony** - Market with only one buyer.
- **market power** - Ability of a seller or buyer to affect the price of a good.

# Average Revenue and Marginal Revenue

边际收入

**marginal revenue** - Change in revenue resulting from a one-unit increase in output.

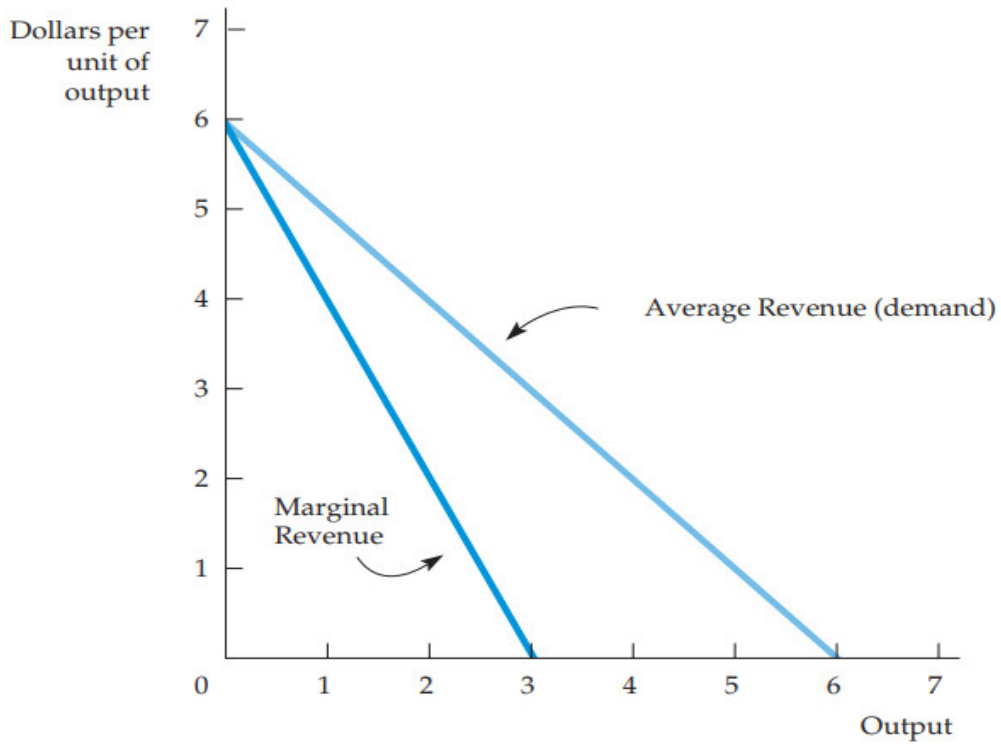
Consider a firm facing the demand curve: $P = 6 - Q$

**TABLE 10.1 TOTAL, MARGINAL, AND AVERAGE REVENUE**

| PRICE (P) | QUANTITY (Q) | TOTAL REVENUE (R) | MARGINAL REVENUE (MR) | AVERAGE REVENUE (AR) |
|---|---|---|---|---|
| $6 | 0 | $0 | — | — |
| 5 | 1 | 5 | $5 | $5 |
| 4 | 2 | 8 | 3 | 4 |
| 3 | 3 | 9 | 1 | 3 |
| 2 | 4 | 8 | –1 | 2 |
| 1 | 5 | 5 | –3 | 1 |

# Average Revenue and Marginal Revenue (Con't)

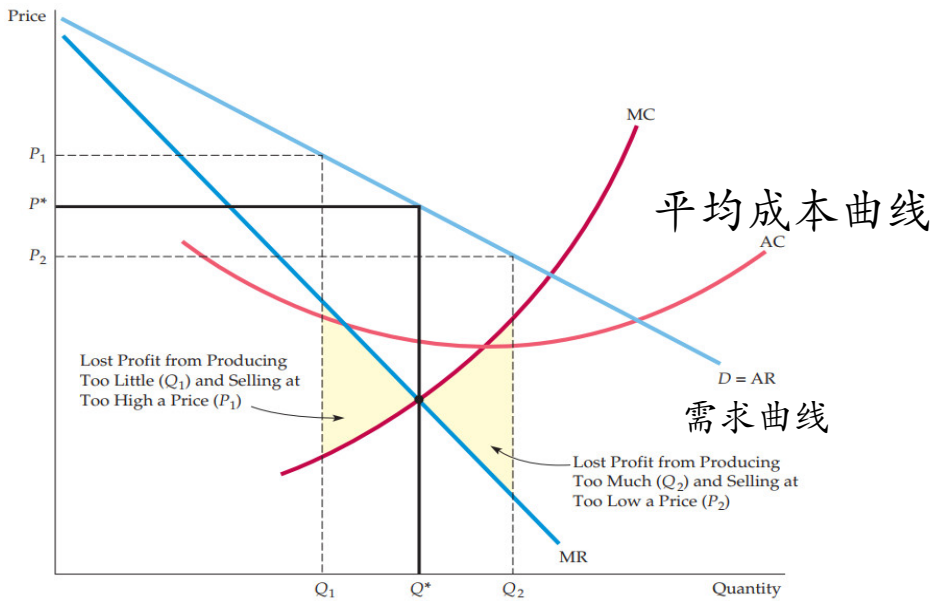
Average and marginal revenue are shown for the demand curve $P = 6 - Q$:



# The Monopolist's Output Decision

**Profit is maximized when marginal revenue equals marginal cost**

- $Q^*$ is the output level at which MR = MC.
- If the firm produces a small output - say, $Q_1$ - it sacrifices some profit.
- Expanding output from $Q^*$ to $Q_2$ would reduce profit.




需求曲线在垄断市场中也代表平均收益（AR, Average Revenue），因为每单位产品的收益就是它的售价。

# Profit Maximization

We can also see algebraically that $Q^*$ maximizes profit. Profit $\Pi$ is the difference between revenue and cost, both of which depend on $Q$:

$$\Pi(Q) = R(Q) - C(Q)$$

Thus the profit-maximizing $Q$ is such that

$$\Delta\Pi/\Delta Q = \Delta R/\Delta Q - \Delta C/\Delta Q = 0$$

Then the profit-maximizing condition is that

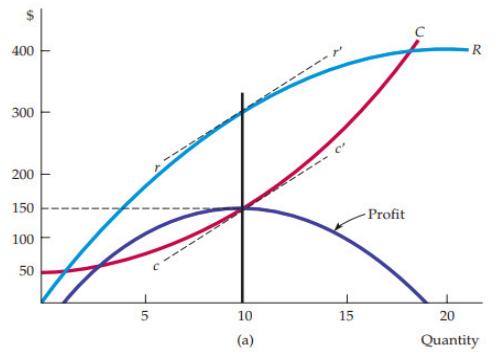
$$MR - MC = 0, \text{or } MR = MC$$

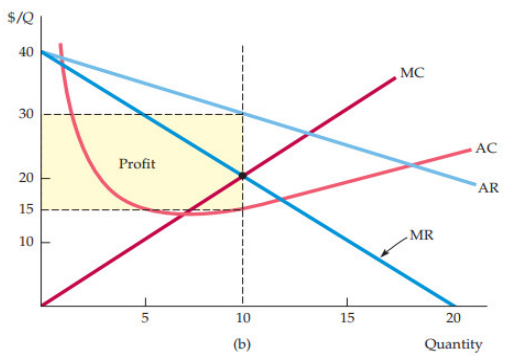
# Example of Profit Maximization

Cost of production: $C(Q) = 50 + Q^2$

Demand: $P(Q) = 40 - Q$

- The profit-maximizing output is $Q^* = 10$, the point where marginal revenue equals marginal cost.
- At this output level, the slope of the profit curve is zero, and the slopes of the total revenue and total cost curves are equal.
- The profit per unit is \$15 and the total profit is \$150.



# A Rule of Thumb for Pricing

With limited knowledge of average and marginal revenue, we can derive a rule of thumb that can be more easily applied in practice. First, write the expression for marginal revenue:

$$MR = \frac{\Delta R}{\Delta Q} = \frac{\Delta(PQ)}{\Delta Q}$$

Then

$$MR = P + Q\frac{\Delta P}{\Delta Q} = P + P\left(\frac{Q}{P}\right)\left(\frac{\Delta P}{\Delta Q}\right)$$

# A Rule of Thumb for Pricing (Con't)

$(Q/P)(\Delta P/\Delta Q)$ is the reciprocal of the elasticity of demand, $1/E_d$. Thus

$$MR = P + P(1/E_d) = MC$$

which can be rearranged to give us

$$\frac{P - MC}{P} = -\frac{1}{E_d}$$

Equivalently, we can rearrange this equation to express price directly as a markup over marginal cost:

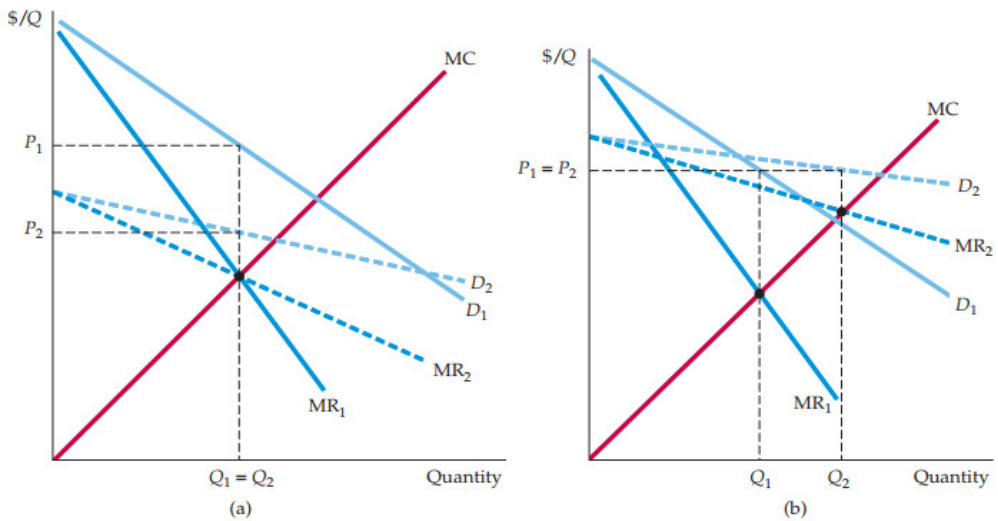
$$P = \frac{MC}{1 + (1/E_d)}$$

# Shifts in Demand

- A monopolistic market has no supply curve. In other words, there is no one-to-one relationship between price and the quantity produced.
- Shifts in demand can lead to changes in price with no change in output, changes in output with no change in price, or changes in both price and output.
- A competitive industry supplies a specific quantity at every price. No such relationship exists for a monopolist, which, depending on how demand shifts, might supply several different quantities at the same price, or the same quantity at different prices.

# Shifts in Demand (Con't)

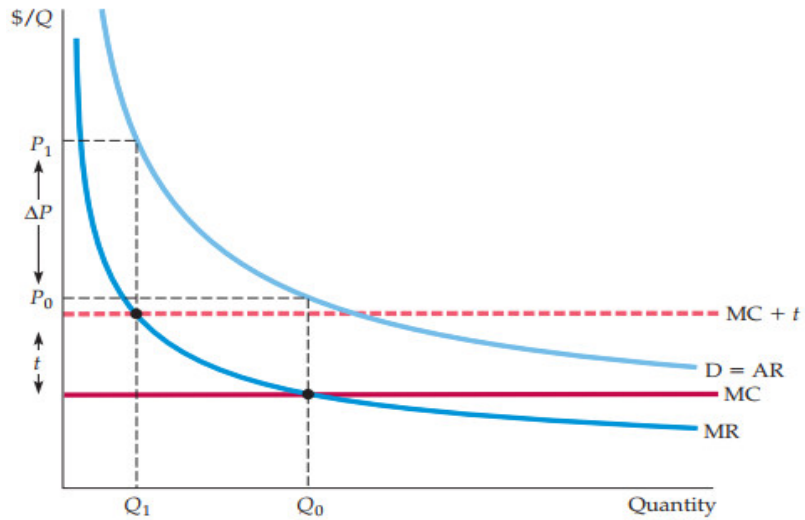
- In **(a)**, the demand curve $D_1$ shifts to new demand curve $D_2$.
- But the new marginal revenue curve $MR_2$ intersects marginal cost at the same point as the old marginal revenue curve $MR_1$.
- The profit-maximizing output remains the same, although price falls from $P_1$ to $P_2$.
- In **(b)**, the new marginal revenue curve $MR_2$ intersects marginal cost at a higher output level $Q_2$.
- But because demand is now more elastic, price remains the same.



# The Effect of a Tax

Suppose a specific tax of $t$ dollars per unit is levied, so that the monopolist must remit $t$ dollars to the government for every unit it sells. If MC was the firm's original marginal cost, its optimal production decision is now given by

$$MR = MC + t$$




In this example, the increase in price $\Delta P$ is larger than the tax $t$.

# The Multiplant Firm

Suppose a firm has two plants. What should its total output be, and how much of that output should each plant produce? We can find the answer intuitively in two steps.

1. Whatever the total output, it should be divided between the two plants so that *marginal cost is the same in each plant*. Otherwise, the firm could reduce its costs and increase its profit by reallocating production.
2. The total output must be such that *marginal revenue equals marginal cost*. Otherwise, the firm could increase its profit by raising or lowering total output.

# The Multiplant Firm (Con't)

Let $Q_1$ and $C_1$ be the output and cost of production for Plant 1, $Q_2$ and $C_2$ be the output and cost of production for Plant 2, and $Q_T = Q_1 + Q_2$ be total output. Then profit is

$$\Pi = PQ_T - C_1(Q_1) - C_2(Q_2)$$

Then

$$\frac{\Delta \Pi}{\Delta Q_1} = \frac{\Delta (PQ_T)}{\Delta Q_1} - \frac{\Delta C_1}{\Delta Q_1} = 0$$

We thus have

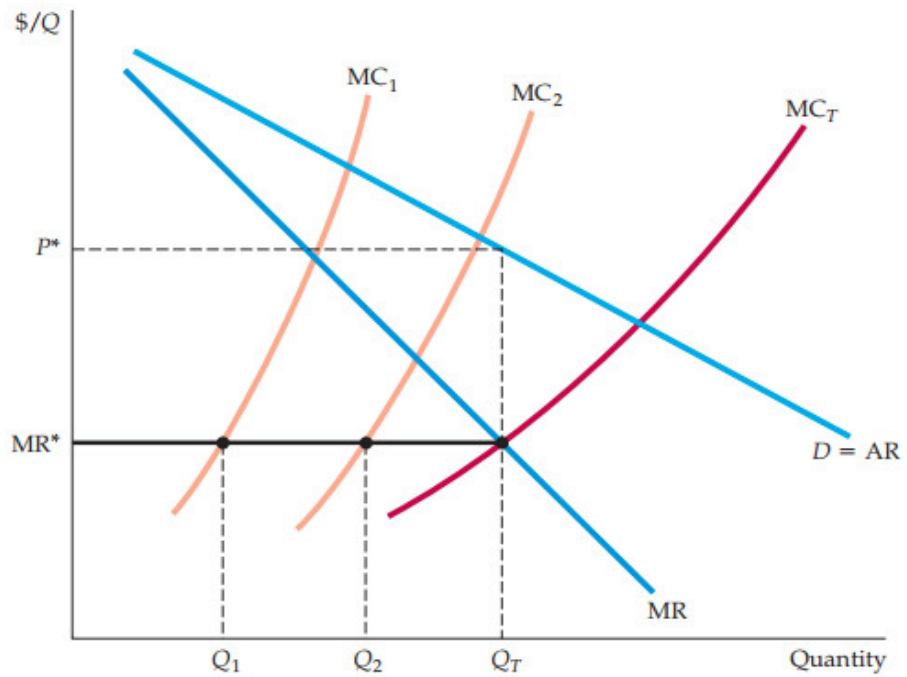
$$MR = MC_1$$

Similarly, we have $MR = MC_2$. Then the firm should produce so that

$$MR = MC_1 = MC_2$$

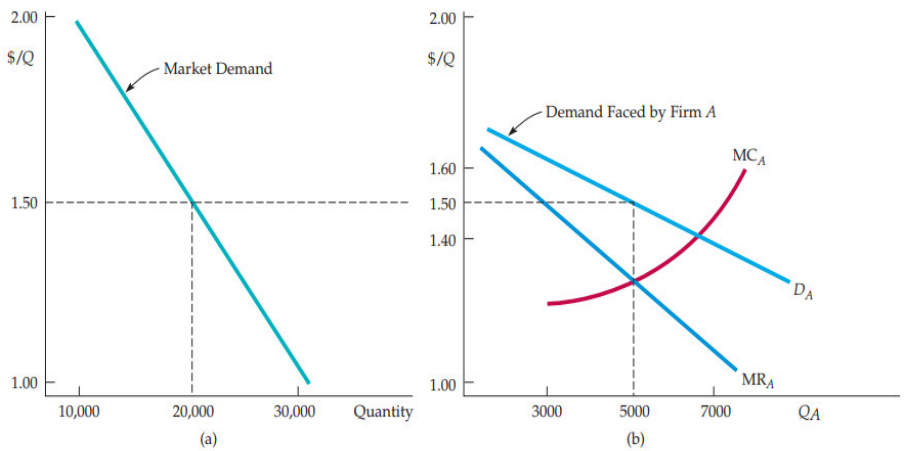
# Producing with Two Plants

A firm with two plants maximizes profits by choosing output levels $Q_1$ and $Q_2$ so that marginal revenue $MR$ (which depends on total output) equals marginal costs for each plant, $MC_1$ and $MC_2$.



# Monopoly Power

- **(a)** shows the market demand for toothbrushes; **(b)** shows the demand curve of Firm A.
- Firm A is likely to face a demand curve which is more elastic than the market demand curve, but which is not infinitely elastic like the demand curve facing a perfectly competitive firm.
- At a market price of \$1.50, elasticity of market demand is $-1.5$ and Firm A's demand elasticity is $-6$.
- Firm A has some monopoly power: Its profit-maximizing price is \$1.50, which exceeds marginal cost.



# Measuring Monopoly Power

The important distinction between a perfectly competitive firm and a firm with monopoly power is: *For the competitive firm, price equals marginal cost; for the firm with monopoly power, price exceeds marginal cost.*

**Lerner Index of Monopoly Power** - Measure of monopoly power calculated as excess of price over marginal cost as a fraction of price.

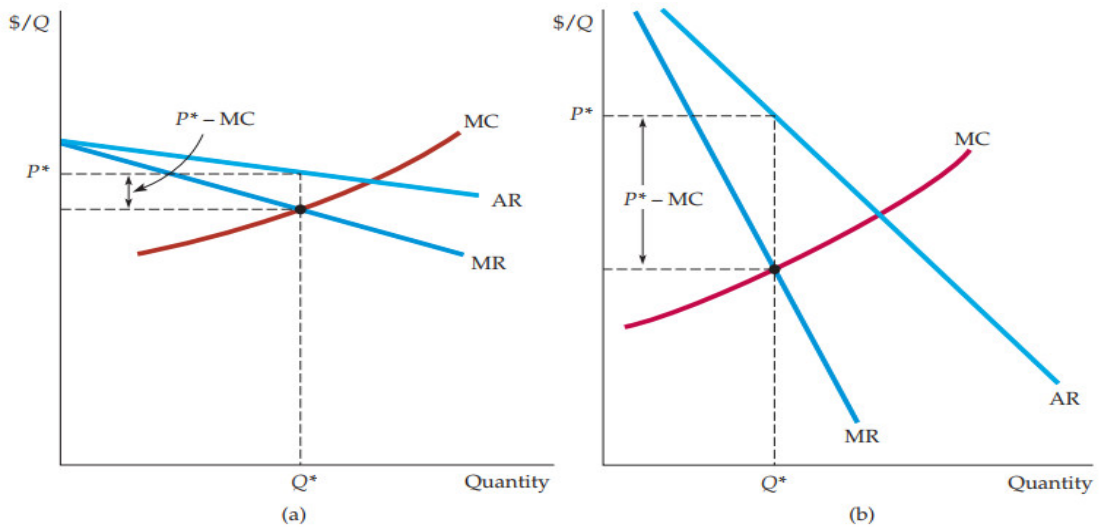
$$L = (P - MC)/P$$

This index can also be expressed as

$$L = (P - MC)/P = -1/E_d$$

# The Rule of Thumb for Pricing

$$P = \frac{MC}{1+(1/E_d)}$$




- If the firm's demand is elastic, as in (a), the markup is small and the firm has little monopoly power.
- The opposite is true if demand is relatively inelastic, as in (b).

# Example: Markup Pricing

- Although the elasticity of market demand for food is small (about $-1$), no single supermarket can raise its prices very much without losing customers to other stores.
- The elasticity of demand for any one supermarket is often as large as $-10$. Thus $P = MC/(1-0.1) = (1.11)MC$.
- Small convenience stores typically charge higher prices because its customers are generally less price sensitive.
- Because the elasticity of demand for a convenience store is about $-5$, the markup equation implies that its prices should be about 25 percent above marginal cost.
- With designer jeans, demand elasticities in the range of $-2$ to $-3$ are typical. This means that price should be 50 to 100 percent higher than marginal cost.

# Sources of Monopoly Power

The ultimate determinant of monopoly power is the firm's elasticity of demand. Three factors determine a firm's elasticity of demand.

1. **The elasticity of market demand.** Because the firm's own demand will be at least as elastic as market demand, the elasticity of market demand limits the potential for monopoly power.
2. **The number of firms in the market.** If there are many firms, it is unlikely that any one firm will be able to affect price significantly.
3. **The interaction among firms.** Even if only two or three firms are in the market, each firm will be unable to profitably raise price very much if the rivalry among them is aggressive, with each firm trying to capture as much of the market as it can.

# The Elasticity of Market Demand

- If there is only one firm - a pure monopolist - its demand curve is the market demand curve. In this case, the firm's degree of monopoly power depends completely on the elasticity of market demand.
- When several firms compete with one another, the elasticity of market demand sets a lower limit on the magnitude of the elasticity of demand for each firm.

# The Number of Firms

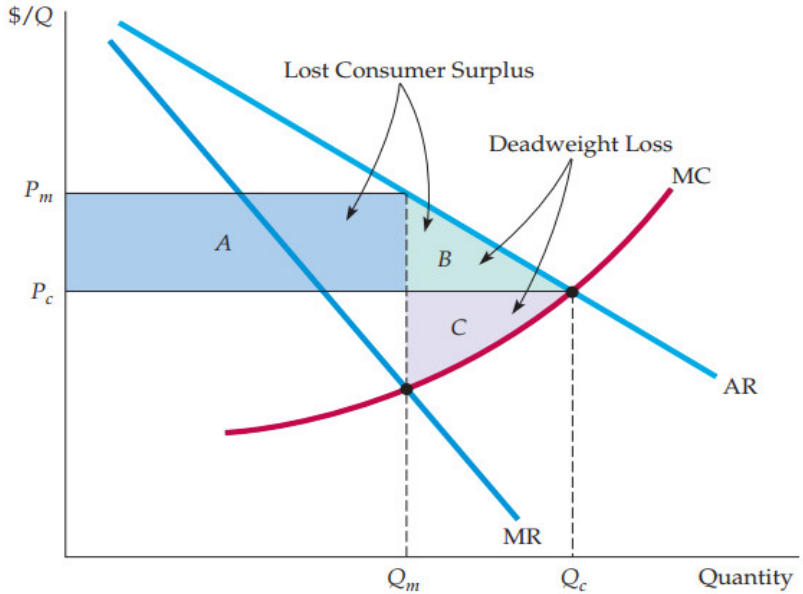
- Other things being equal, the monopoly power of each firm will fall as the number of firms increases.
- When only a few firms account for most of the sales in a market, we say that the market is highly *concentrated*.
- **barrier to entry** - Condition that impedes entry by new competitors.
    - Sometimes there are natural barriers to entry: Patents, copyrights, and licenses
- In some cases, *economies of scale* may be so large that it is most efficient for a single firm - *a natural monopoly* - to supply the entire market.

# The Interaction Among Firms

- Other things being equal, monopoly power is smaller when firms compete aggressively and is larger when they cooperate.
  - Because raising prices in concert rather than individually is more likely to be profitable, collusion can generate substantial monopoly power.
- A firm's monopoly power often changes over time, as its operating conditions (market demand and cost), its behavior, and the behavior of its competitors change. Monopoly power must therefore be thought of in a dynamic context.
- Real or potential monopoly power in the short run can make an industry more competitive in the long run: Large short-run profits can induce new firms to enter an industry, thereby reducing monopoly power over the longer term.

# Deadweight Loss from Monopoly Power

- The shaded rectangle and triangles show changes in consumer and producer surplus when moving from competitive price and quantity, $P_c$ and $Q_c$, to a monopolist's price and quantity, $P_m$ and $Q_m$.
- Because of the higher price, consumers lose $A+B$ and producer gains $A-C$. The deadweight loss is $B+C$.

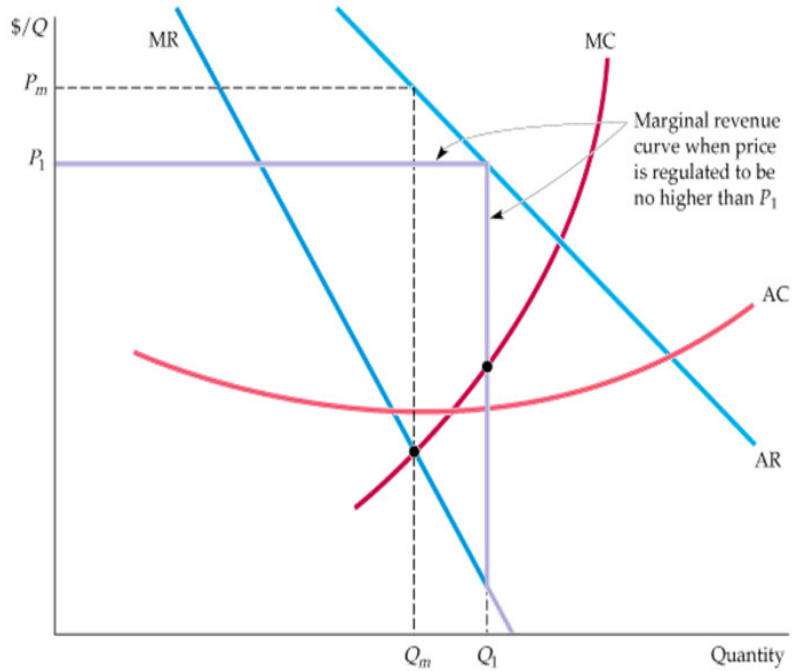



**rent seeking** - Spending money in socially unproductive efforts to acquire, maintain, or exercise monopoly.

- We would expect the economic incentive to incur rent-seeking costs to bear a direct relation to the gains from monopoly power (i.e., rectangle A minus triangle C.)

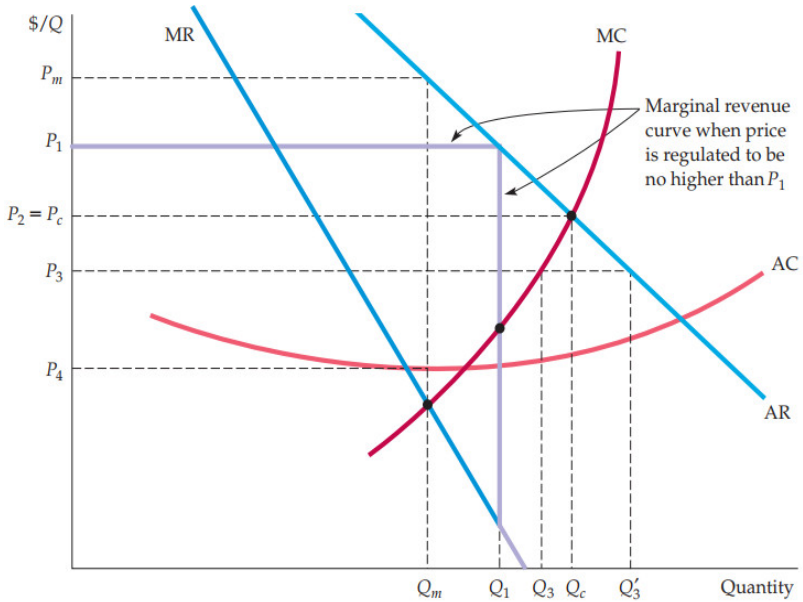
# Price Regulation

- If left alone, a monopolist produces $Q_m$ and charges $P_m$.
- When the government imposes a price ceiling of $P_1$ the firm's average and marginal revenue are constant and equal to $P_1$ for output levels up to $Q_1$.
- For larger output levels, the original average and marginal revenue curves apply.
- The new marginal revenue curve is the dark purple line, which intersects the marginal cost curve at $Q_1$.



# Price Regulation (Con't)

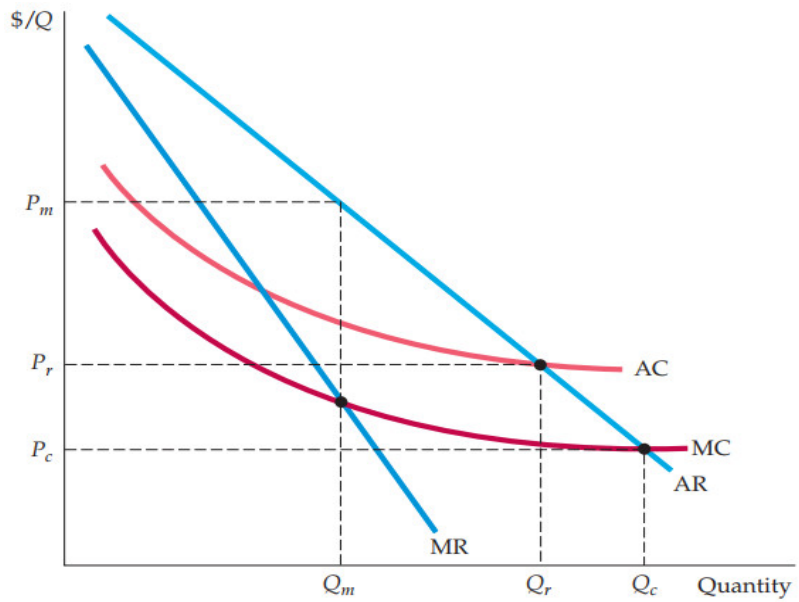
- When price is lowered to $P_c$, at the point where marginal cost intersects average revenue, output increases to its maximum $Q_c$. This is the output that would be produced by a competitive industry.
- Lowering price further, to $P_3$, reduces output to $Q_3$ and causes a shortage, $Q_3' - Q_3$.



# Natural Monopoly

**natural monopoly** - Firm that can produce the entire output of the market at a cost lower than what it would be if there were several firms.

- A firm is a natural monopoly because it has economies of scale (declining average and marginal costs) over its entire output range.
- If price were regulated to be $P_c$ the firm would lose money and go out of business.
- Setting the price at $P_r$ yields the largest possible output consistent with the firm's remaining in business; excess profit is zero.



# Regulation in Practice

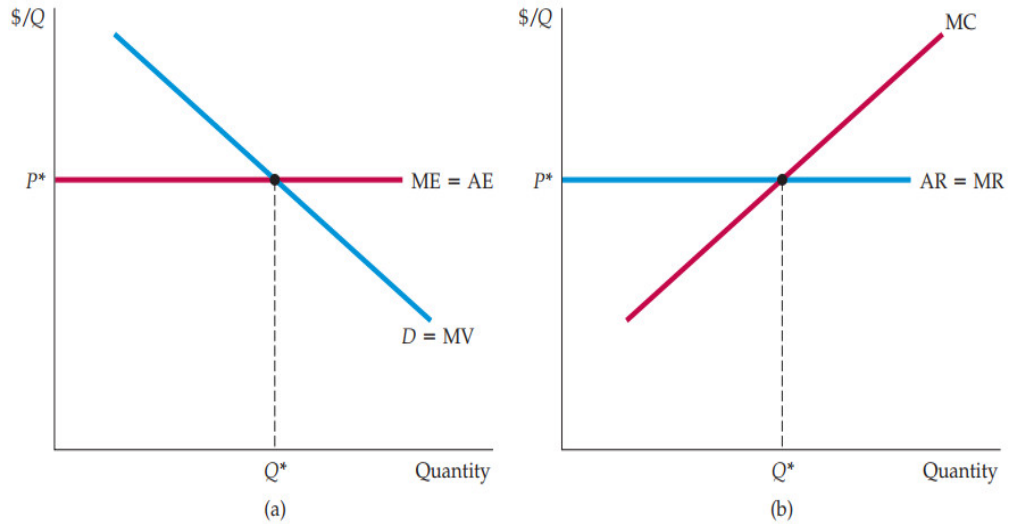
- **rate-of-return regulation** - Maximum price allowed by a regulatory agency is based on the (expected) rate of return that a firm will earn.
- Another approach to regulation is setting price caps based on the firm's variable costs. A price cap can allow for more flexibility.
  - For example, a firm could raise its prices each year (without having to get approval from the regulatory agency) by an amount equal to the actual rate of inflation, minus expected productivity growth.

# Monopsony

- **monopsony power** - Buyer's ability to affect the price of a good.
- **oligopsony** - Market with only a few buyers.
- **marginal value** - Additional benefit derived from purchasing one more unit of a good.
- **marginal expenditure** - Additional cost of buying one more unit of a good.
- **average expenditure** - Price paid per unit of a good.

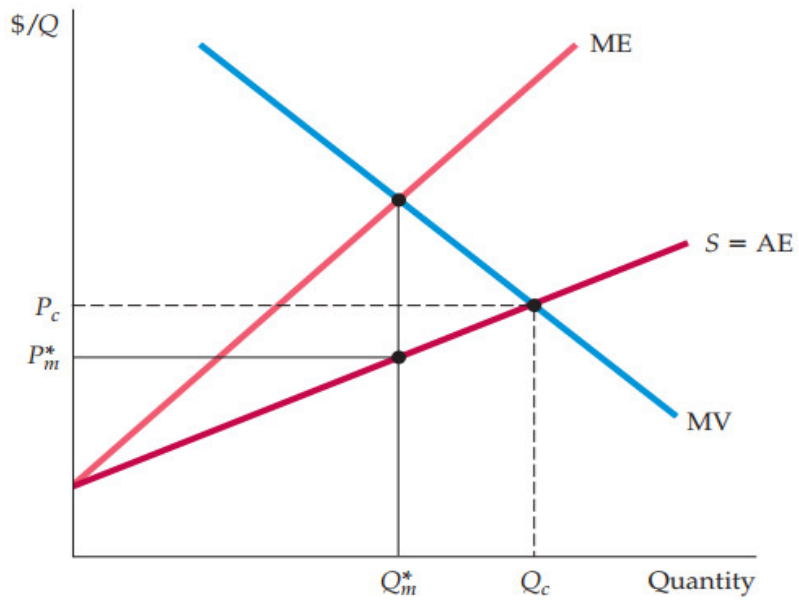
# Competitive Buyer Compared to Competitive Seller

- In (a), the competitive buyer takes market price $P^*$ as given. Therefore, marginal expenditure and average expenditure are constant and equal; quantity purchased is found by equating price to marginal value (demand).
- In (b), the competitive seller also takes price as given. Marginal revenue and average revenue are constant and equal; quantity sold is found by equating price to marginal cost.



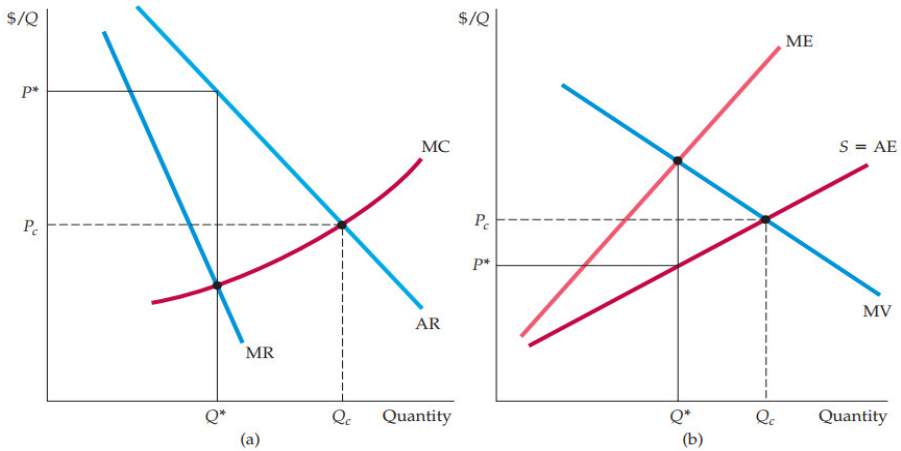
# Monopsonist Buyer

- The market supply curve is monopsonist's average expenditure curve $AE$.
- Because average expenditure is rising, marginal expenditure lies above it.
- The monopsonist purchases quantity $Q_m^*$ and pays price $P_m^*$ per unit.
- In a competitive market, price and quantity, $P_c$ and $Q_c$, are both higher. They are found at the point where average expenditure (supply) and marginal value (demand) intersect.



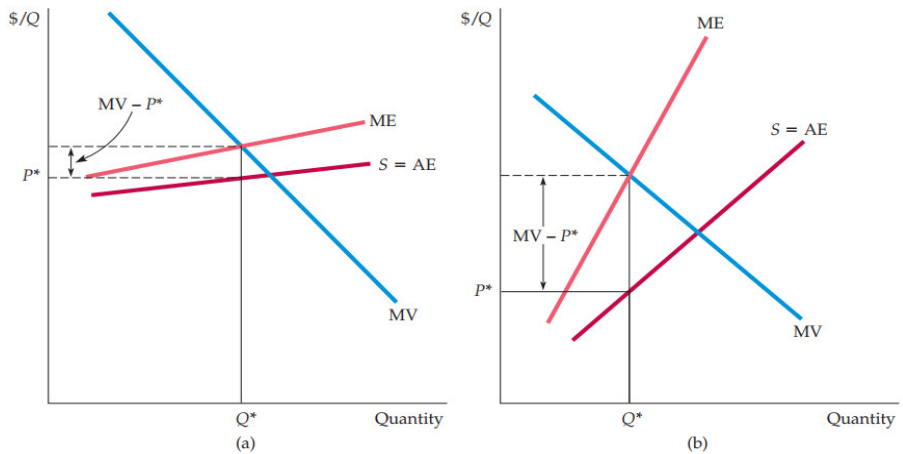
# Monopsony and Monopoly Compared

- **(a)** The monopolist produces where marginal revenue intersects marginal cost.
  - Average revenue exceeds marginal revenue, so that price exceeds marginal cost.
- **(b)** The monopsonist purchases up to the point where marginal expenditure intersects marginal value.
  - Marginal expenditure exceeds average expenditure, so that marginal value exceeds price.



# Monopsony Power: Elastic versus Inelastic Supply

- Monopsony power depends on the elasticity of supply.
- When supply is elastic, as in **(a)**, marginal expenditure and average expenditure do not differ by much, so price is close to what it would be in a competitive market.
- The opposite is true when supply is inelastic, as in **(b)**.

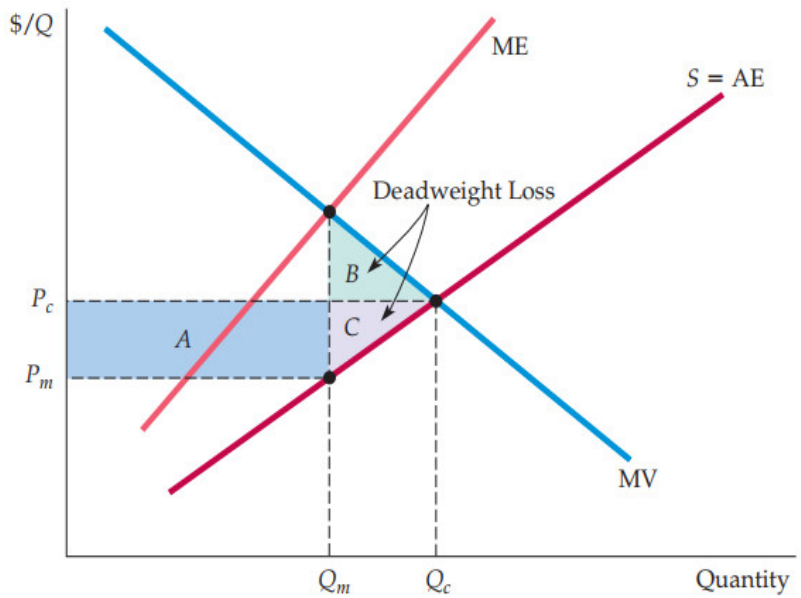


# Sources of Monopsony Power

- **Elasticity of Market Supply**
  - If only one buyer is in the market - a pure monopsonist - its monopsony power is completely determined by the elasticity of market supply. If supply is highly elastic, monopsony power is small and there is little gain in being the only buyer.
- **Number of Buyers**
  - When the number of buyers is very large, no single buyer can have much influence over price. Thus each buyer faces an extremely elastic supply curve, so that the market is almost completely competitive.
- **Interaction among buyers**
  - If four buyers in a market compete aggressively, they will bid up the price close to their marginal value of the product, and will thus have little monopsony power. On the other hand, if those buyers compete less aggressively, or even collude, prices will not be bid up very much, and the buyers' degree of monopsony power might be nearly as high as if there were only one buyer.

# The Social Costs of Monopsony Power

- The shaded rectangle and triangles show changes in buyer and seller surplus when moving from competitive price and quantity, $P_c$ and $Q_c$, to the monopsonist's price and quantity, $P_m$ and $Q_m$.
- Because both price and quantity are lower, there is an increase in buyer (consumer) surplus given by $A - B$.
- Producer surplus falls by $A + C$, so there is a deadweight loss given by triangles $B$ and $C$.



# Bilateral Monopoly

**bilateral monopoly** - Market with only one seller and one buyer.

- It is difficult to predict the price and quantity in a bilateral monopoly. Both the buyer and the seller are in a bargaining situation.
- Although bargaining may still be involved, we can apply a rough principle here: *Monopsony power and monopoly power will tend to counteract each other.* In other words, the monopsony power of buyers will reduce the effective monopoly power of sellers, and vice versa.
- This tendency does not mean that the market will end up looking perfectly competitive, but in general, monopsony power will push price closer to marginal cost, and monopoly power will push price closer to marginal value.

# Summary

- Monopoly
  - Profit is maximized when $MR = MC$.
  - Rule of thumb for pricing

$$\frac{P - MC}{P} = -\frac{1}{E_d} \text{ or } P = \frac{MC}{1 + (1/E_d)}$$

  - Effect of tax: $MR = MC + t$
  - Multiplant firm: $MR = MC_1 = MC_2$
- Monopoly power
  - Lerner index of monopoly power

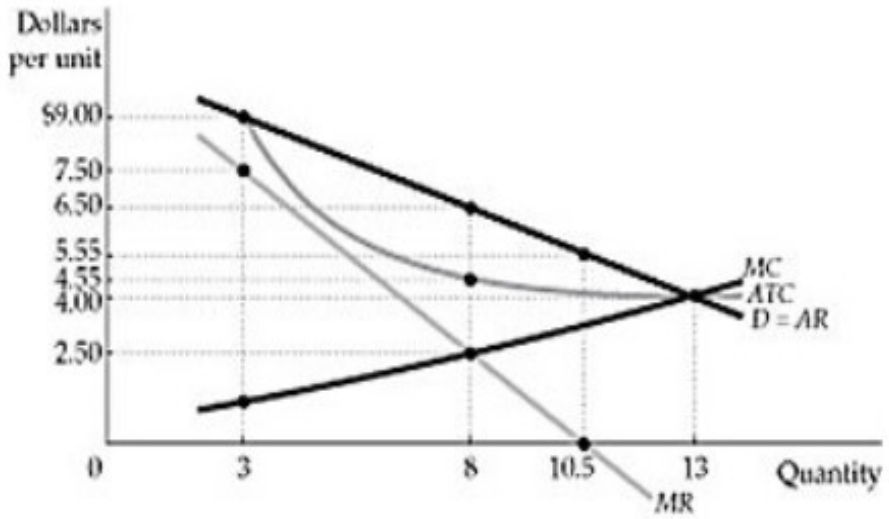
$$L = (P - MC)/P = -1/E_d$$

  - Three sources of monopoly power

# Summary (Con't)

- Social costs of monopoly power
  - deadweight loss from monopoly power
  - rent seeking, price regulation, natural monopoly
- Monopsony and Monopsony power
  - monopsonist buyer
  - comparison between monopsony and monopoly
  - Monopsony power depends on the elasticity of supply.
  - Three sources of monopsony power
  - social costs of monopsony power

# Exercises

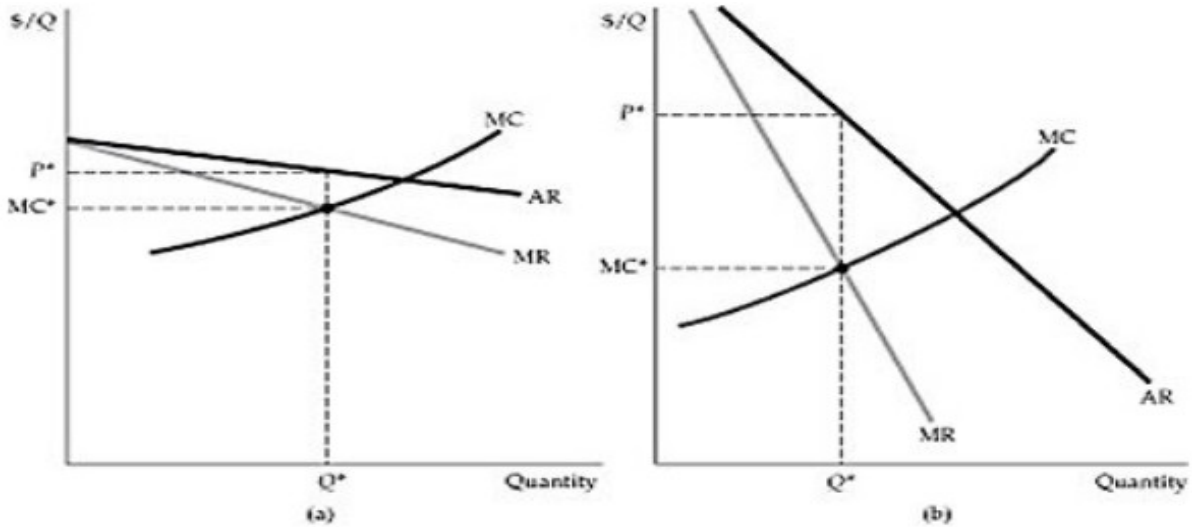



1. When the monopoly maximizes profit, how much is the amount of profit?
   a. $4.50
   b. $14.7
   c. $15.6
   d. $52.00

# Exercises (Con't)

2. If a monopolist sets her output such that marginal revenue, marginal cost and average total cost are equal, economic profit must be:
   a. negative.
   b. positive.
   c. zero.
   d. indeterminate from the given information.

# Exercises (Con't)



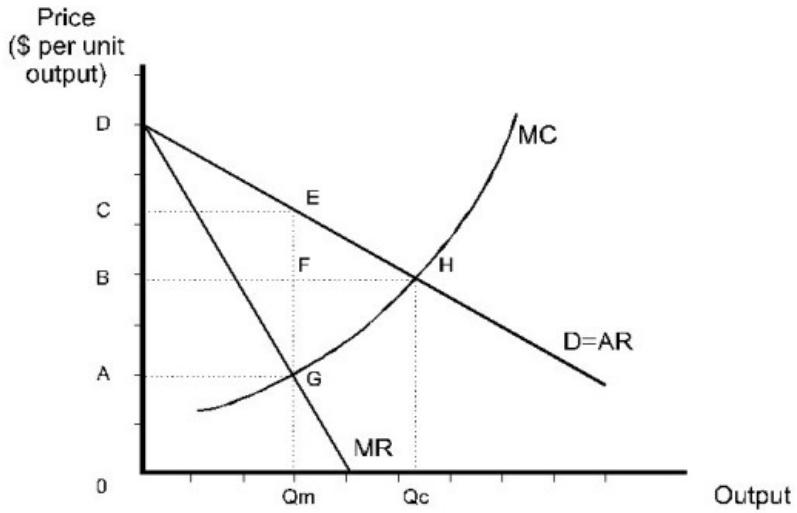

3. Which monopoly has greater monopoly power?
   a. The monopoly in panel (a).
   b. The monopoly in panel (b).
   c. Both firms have the same market power.
   d. Cannot tell from this figure.

# Exercises (Con't)

4. What is the value of the Lerner index under perfect competition?
   a. 1
   b. 0
   c. infinity
   d. two times the price

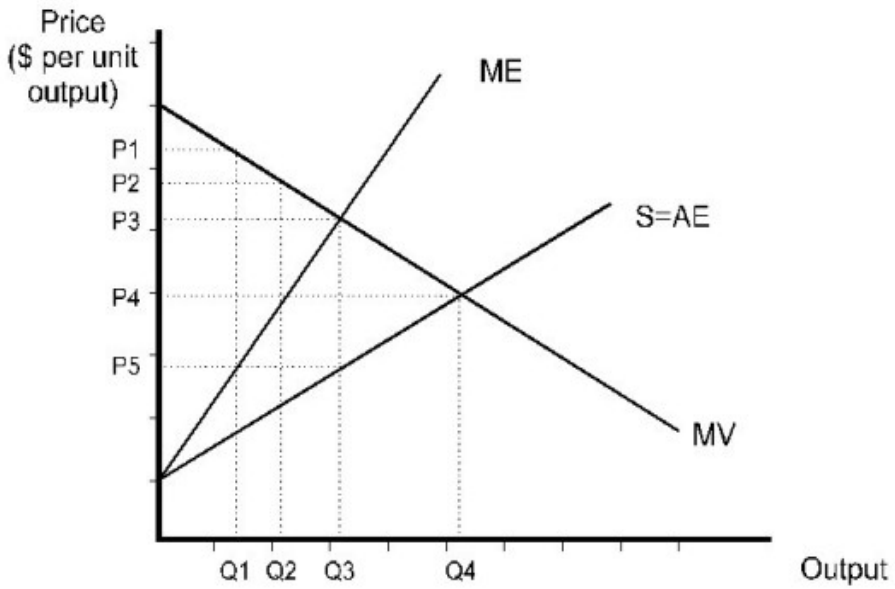
5. Which of the following is NOT associated with a high degree of monopoly power?
   a. A relatively inelastic demand curve for the firm
   b. A small number of firms in the market
   c. Significant price competition among firms in the market
   d. Significant barriers to entry

# Exercises (Con't)



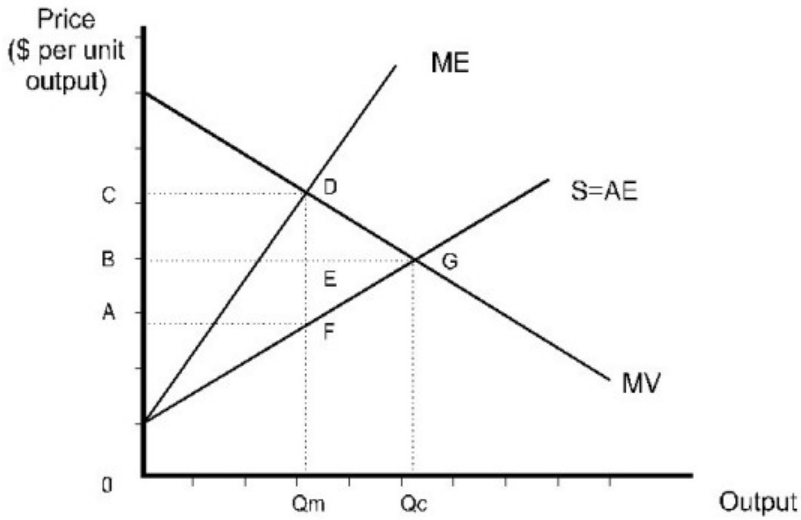

6. In moving from the competitive level of output and price to the monopoly level of output and price, the monopolist is able to add to producer surplus:
   a. the area BCEF.
   b. the area BCEF less the area GFH.
   c. the area BCEH.
   d. the area BCEH less the area GFH.
   e. none of the above

# Exercises (Con't)




7. What price will the monopsonist pay when maximizing profit?
   a. P1
   b. P2
   c. P3
   d. P4
   e. P5

# Exercises (Con't)




8. The above diagram shows marginal value and expenditure curves for a monopsony. In moving from the competitive price and quantity to the monopsony price and quantity, the deadweight loss from monopsony power is the area:
   a. ACDF
   b. CDE
   c. EDG
   d. FDG
   e. BCDG

# Exercises (Con't)

9. The monopsonist's markdown in the buying price increases as:
   1. the supply elasticity declines.
   2. the supply elasticity increases.
   3. the demand elasticity declines.
   4. the demand elasticity increases.